Rithish S

Programmer Analyst Intern | CS Fundamentals, AWS (S3, DynamoDB, EC2), Schedulers & Workflows
+91 99522 52304 | rithishcodespace@gmail.com | LinkedIn | GitHub | LeetCode

Education

Bannari Amman Institute of Technology — Sathyamangalam, Tamil Nadu
Bachelor of Engineering in Computer Science and Engineering — *Aug. 2024 – May 2028*
CGPA: **8.28/10**

Technical Skills

Languages: Java, C++, Python, SQL, JavaScript, TypeScript
Computer Science: Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks
Cloud & AWS: AWS (S3, DynamoDB, EC2), Docker, Kubernetes, Linux/POSIX, Git, GitHub Actions, CI/CD
Web Services & Architecture: REST APIs, Microservices, Schedulers, Background Workflows, State Machines, Node.js, Fastify, Clean Architecture
Databases: PostgreSQL, MySQL, DynamoDB, Redis, Prisma ORM
Testing & Quality: Automated Unit Testing, Integration Testing, System Troubleshooting, Debugging, Regression Testing

Open Source

ESLint Core | *JavaScript, AST, Static Analysis, Troubleshooting* PR #21218

- Diagnosed and resolved a subtle floating-point precision bug in ESLint's `no-loss-of-precision` static analysis rule, correctly distinguishing literal zero values from decimal numbers that underflow under IEEE 754 binary64 specifications.
- Interpreted complex AST codebase with minimal guidance; engineered comprehensive regression test suites covering signed zero, subnormal values, and non-decimals, passing **43 automated CI verification checks** prior to production merge.

Projects

DBVault – Database Backup & Scheduler Tool | *TypeScript, Node.js, Redis, BullMQ, AWS S3, Docker* GitHub

- Engineered a database backup, restore, and scheduling tool supporting **PostgreSQL, MySQL, MongoDB, and SQLite** with automated cron scheduling, stream compression, and SHA-256 integrity validation.
- Designed resilient background workflows and job state machines with **Redis and BullMQ**, implementing worker concurrency, rate limiting, exponential backoff retries, and distributed locking to prevent duplicate execution.
- Built pluggable storage adapters for automated offloading to **AWS S3** object storage and AES-256-GCM encryption; authored **120+ automated unit and integration tests** validating high reliability and zero data corruption.

PatentIQ – AI Research & Service Integration | *Python, Node.js, Fastify, PostgreSQL, REST APIs, Redis* GitHub

- Developed an automated research service utilizing **multi-step workflows, tool state machines, and REST APIs** to extract, analyze, and synthesize technical patent claims from unstructured document repositories.
- Built a modular **Fastify and PostgreSQL** backend utilizing Clean Architecture, dependency injection, and schema validation to orchestrate service-level integrations across vector search indices and processing endpoints.
- Optimized query workflows and caching layers with Redis, reducing latency for complex claim-overlap analysis while preserving modularity for future service integrations and internal tooling.

Experience

Web Developer – Internal Systems — Feb. 2025 – Dec. 2025
Students Special Group — *Student Organization*

- Built and maintained an internal **Project Management Portal** (React, Node.js, MySQL) used across departments to automate task assignment, track milestones, and eliminate manual spreadsheet tracking.
- Acted as a **liaison between academic stakeholders and developers** to elicit, analyze, and validate functional requirements for a **Faculty Reward Points System**, designing relational schemas and reporting APIs.
- Investigated and debugged system bottlenecks across database queries and API endpoints; mentored **50+ junior developers** in software design principles, unit testing, and Git version control.

Achievements & Certifications

- **1200+ DSA Problems Solved** across LeetCode, CodeChef, and GeeksforGeeks, demonstrating mastery in algorithms, data structures, and computational problem-solving.
- **Certifications**: **AWS Cloud Practitioner Essentials** (AWS Cloud Architecture, S3, EC2, DynamoDB), **Oracle Cloud Infrastructure (OCI 2025 Associate)**, **NPTEL Python** (Elite + Topper, Top 5%, 95%).
- **IEEE DevSpark Hackathon – 1st Place** among 120+ teams for architecting **Plantera**, an automated environmental data monitoring system.
- **BIT Hackathon – 1st Place** among 200+ teams by delivering a full-stack **Leave Management System** under rapid development constraints.